Data sources

RESPONSIBLE AI DATA MANAGEMENT

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Coming up..

- Data source types
- Limitations and selection
- Integrating multiple data sources

Why data source is important

- Integrity
- Diversity
- Fair representation

[1] Image by Streamline HQ

Types by origin

- Primary
  - Data collected within the project
  - Compliance and consent
- Secondary
  - Data acquired from existing resources
  - Licensing agreements

Types by nature

- Quantitative
  - Numeric data
- Qualitative
  - Non-numeric
- Mixed
  - Combination of numeric and non-numeric

[1] Image by Streamline HQ

Types by temporality

- Static
  - Does not change over time
  - Census data
  - Corporate addresses
- Dynamic
  - Updated real-time
  - Social media streams
  - API
  - Financial market feeds
  - Sensor data

[1] Image by Streamline HQ

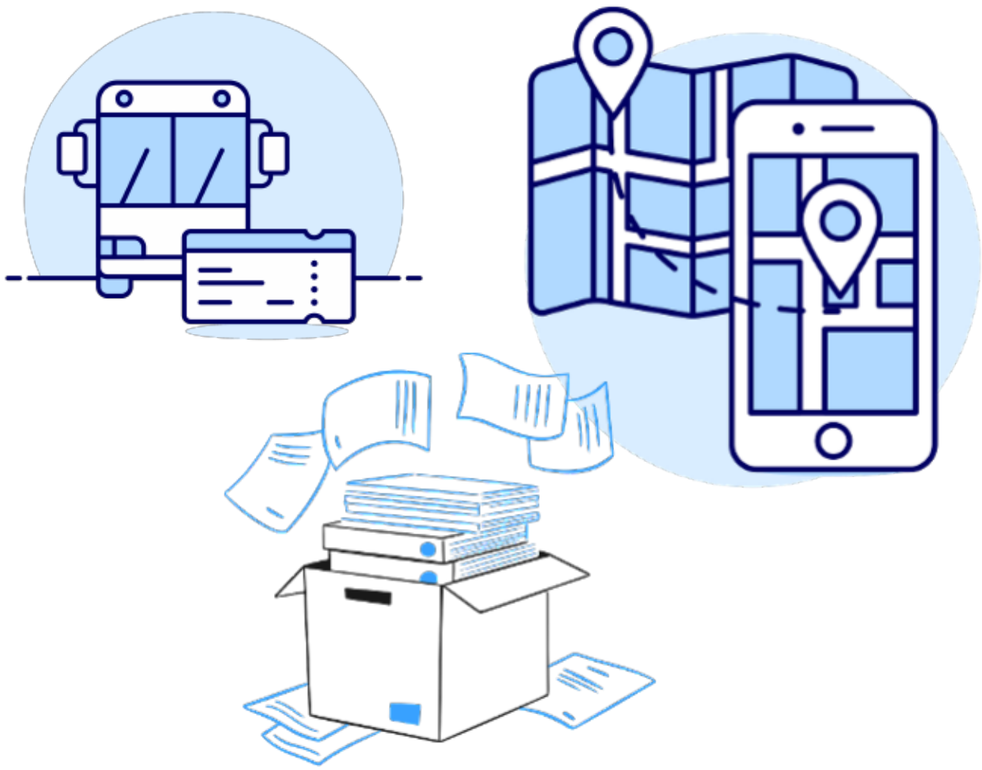
Diversity and fairness in data sources

- Data collectors' direct biases
- Measurable bias checks
- Not accurately representing current realities, outdated biases
- Inherited biases from the original context
- Nuanced analysis
- Continually evolve, possibly introducing real-time biases

Urban traffic flow project

Data sources:

- Historical traffic data
- City council meeting notes
- GPS tracking data



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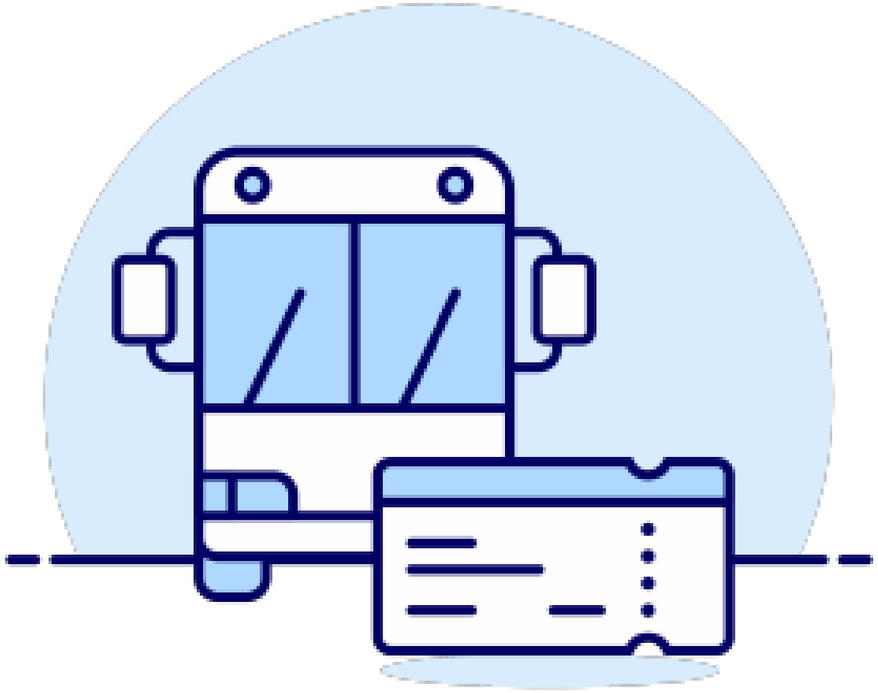
Historical traffic data

- City's transportation department
- Last 5 years
- Includes vehicle counts and time of day/week

This is a **primary static quantitative source**



[1] Image by Streamline HQ

Council meetings minutes

- Public records available at the council website
- Urban planning and traffic management summaries

This is a **qualitative secondary source**

[1] Image by Streamline HQ

GPS data

- Primary dynamic source
- Immediate insights into current traffic conditions, speeds, and delays

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Let's practice!

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Data source limitations

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Data source and common limitations

- Legal compliance
- Bias
- Methodology
- Role of domain knowledge

Legal and access-based limitations

- Restricted use of data for particular types of projects
- Additional compliance
- Prohibitive cost

[1] Image by Streamline HQ

Bias in data sources

- Systematic errors
- Distorted perceptions
- Uneven outcomes
- Disadvantage certain groups

Types of bias

- Historical bias:
  - Irrelevant patterns and outcomes
- Selection bias:
  - Choosing which data points to include
- Sampling bias
  - Method used for sampling

[1] Image by Streamline HQ

Bias and origin-based limitations

- Restricted data coverage
- Cultural and geographical constraints
- Restricted in scope and inclusion

Bias and methodology-based limitations

- Choice of data collection methods
- Sampling approaches

Domain knowledge

- Limitations or hidden biases
- Engage with domain experts early
- Mitigate limitations before the modeling

Urban traffic flow project

Data sources:

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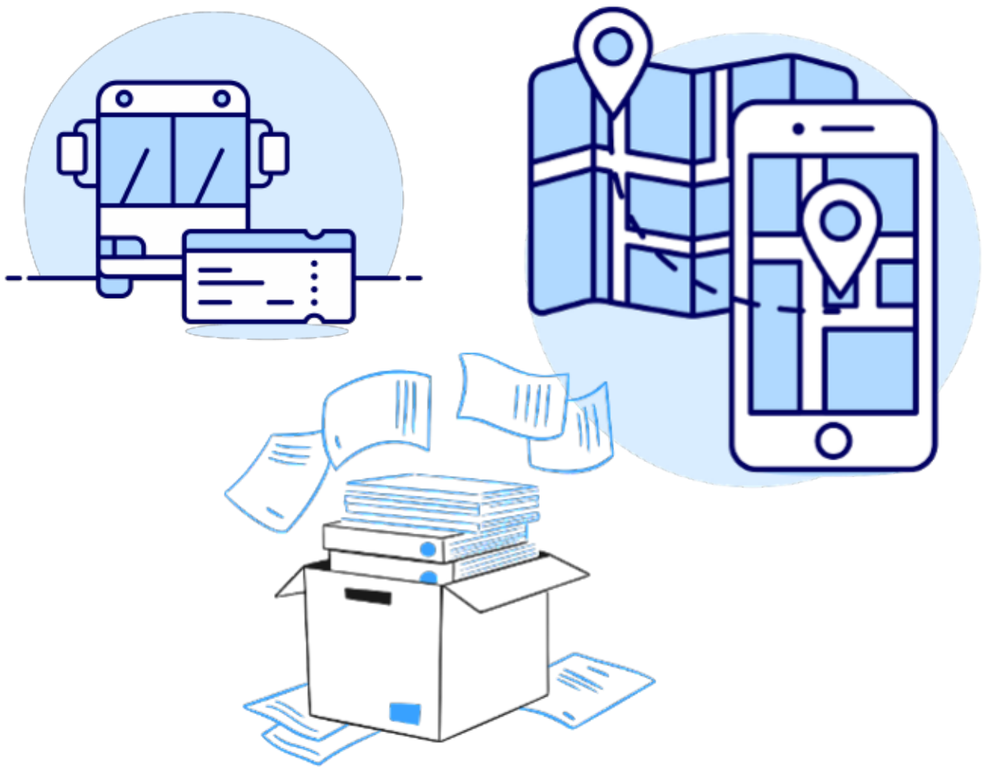
- City council meeting notes
- GPS tracking data



[1] Images by Streamline HQ

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Historical traffic count data:

- Historical bias
- Past urban layouts

Meeting minutes data:

- Selection bias
- Disproportionately representation of some community members

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Urban traffic flow project

GPS Tracking Data:

- Sampling and selection biases
- May not represent all commuters

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Data source selection

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Why select?

- Ensure data quality
- Legal compliance
- Fairness

Step 1. Project relevance

- Relevance to the project objectives
- Check for alignment with
  - Subject area
  - Scope
  - Anticipated outcomes

Step 2. Data source integrity

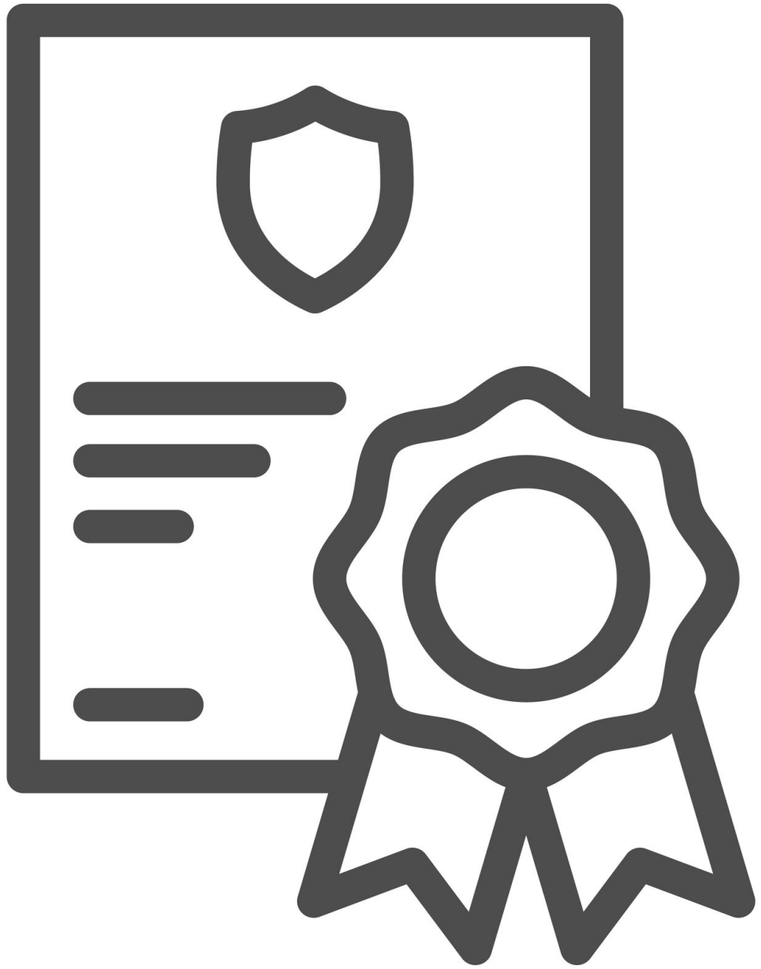
- Assess integrity and trustworthiness
- Reviews and testimonials
- Transparency in data collection
- Compliance and licensing
- High-quality: regular updates

Step 3. Legal compliance

- Lawfulness of data
- Legal compliance for the project
- Legal counsel:
  - Applicable laws and restrictions
  - Data anonymization
  - Data security requirements
- Approved by a legal team

Step 4. Technical quality

- Structural integrity and usability
- Complete
- Consistent
- Accurate
- Timely

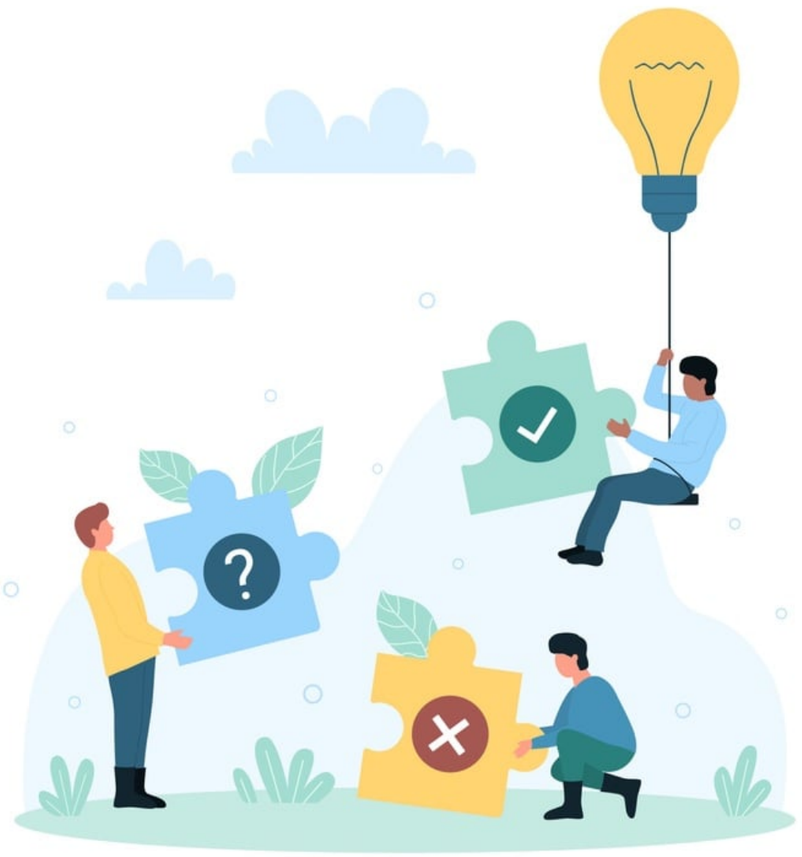


Step 5. Bias and representativeness

- Demographic representation analysis
- Protected characteristics
- Analyze the distribution of groups
- Fairness metrics
- Data augmentation

Step 6. Selection

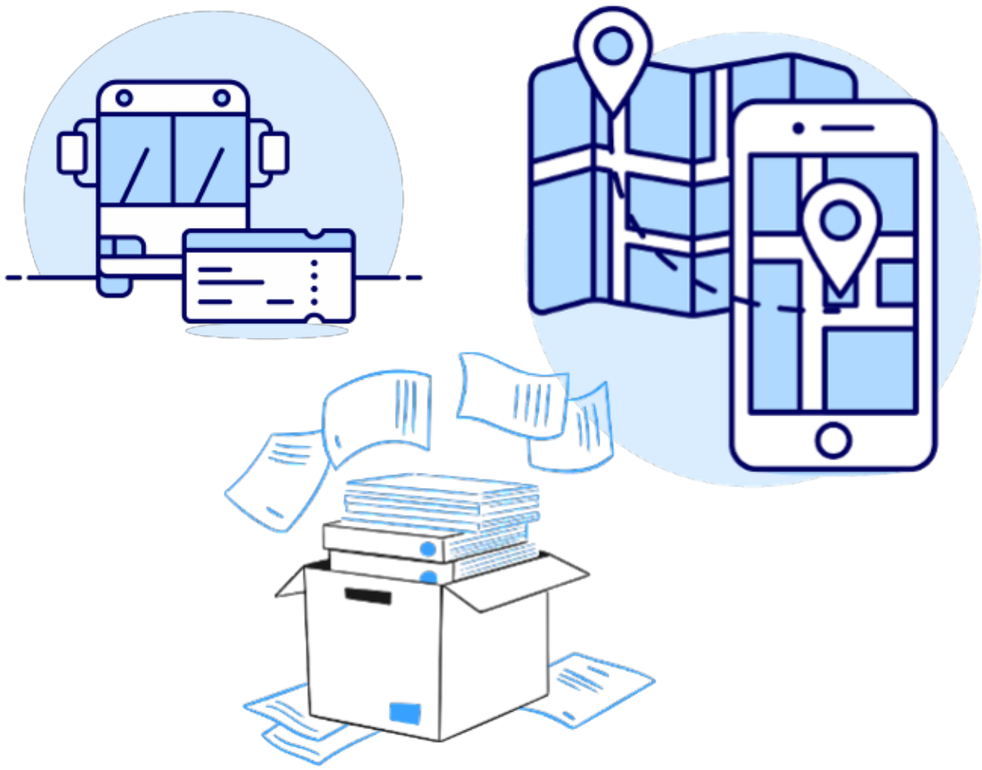
- Include if:
  - Consistently aligns
  - Can be corrected with transformation, augmentation, or algorithms
- Exclude if:
  - Lacks in key areas
- Consult domain experts



Urban traffic flow project

Data sources:

1. Traffic count data
2. Council meeting notes
3. GPS tracking data
4. Social media mentions of traffic conditions
5. Commuter survey data



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Exclude:

- Social media data
- Commuter survey data

Alterations:

- Council meeting notes
- Traffic count data
  - Additional data from sensors
- GPS data
  - Additional data from cameras

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Data integration

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What we will cover

- Why integration is necessary
- Its benefits, and its complications
- Steps for data integration

Why have multiple sources?

- Comprehensive detailed view
- Safety net
- Data diversity and fairness
- Explainability, transparency and accountability

Beware of issues

- Compromise data quality
- Introduce inconsistencies
- Amplify biases
- Reduce representation
- Model complexity
- Reduced transparency and explainability

[1] Image by Streamline HQ

Step 1. Data sources selection

- Follow the evaluation steps
- Assess the data sources
- More balanced and comprehensive dataset

Step 2. Aligning data types

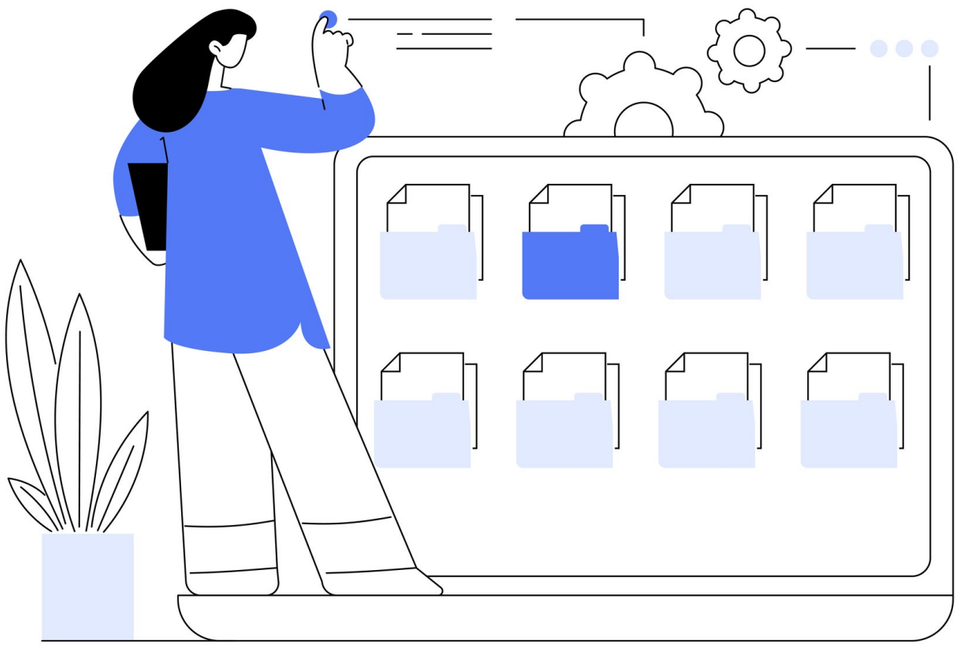
- Identify common variables
- Standardize names and formats
- Normalize numerical data
- Consolidating categorical data
- Align data granularity

Step 3. Bias and representation enhancement

- Weighting
  - Domain knowledge
  - Assign weights to under or overrepresented groups
- Balancing
  - Equal representation
  - Over or undersampling
- Algorithmic checks
- Gap analysis

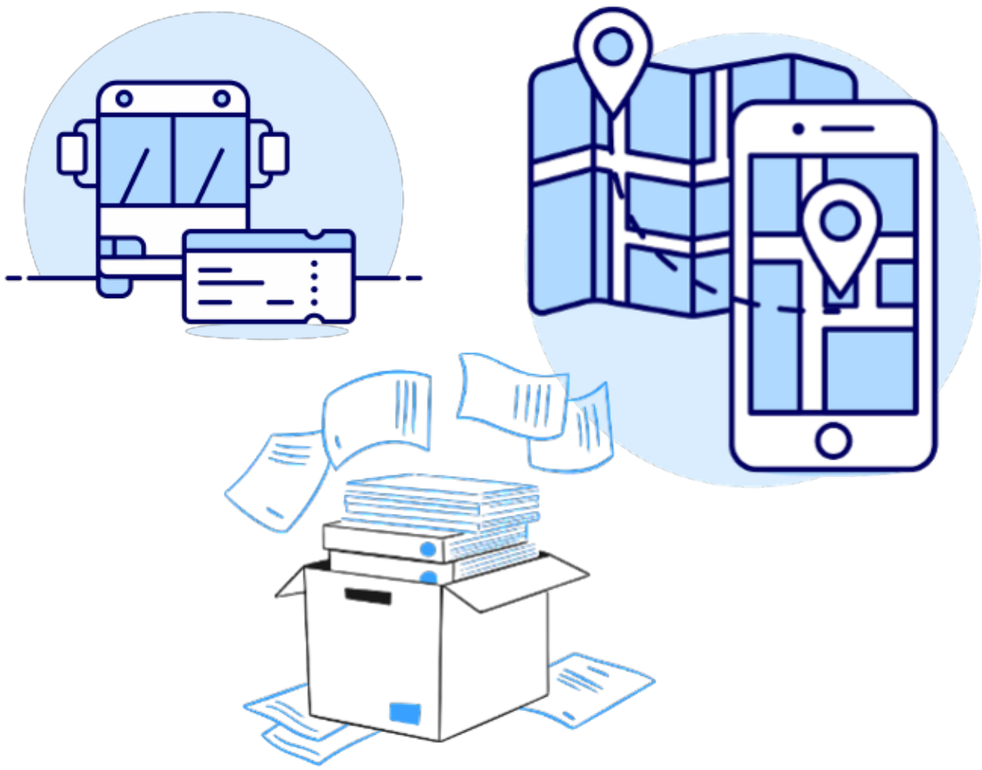
Step 4. Document

- Detailed records:
  - Data integration steps
  - Decisions made
- Detailed metadata:
  - Data sources
  - Collection methodology
  - Applied transformation



Urban traffic flow project

- Select data sources
- Identify common features
- Develop a Unified Data Model



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Urban traffic flow project

- Use statistical techniques for bias and representation
- Apply weighting adjustments
- Gap analysis and reweighting
- Document

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